

Module 11 — Capstone: Pythia8 → Geant4

Supplementary to the sPHENIX Standalone Course

1. The full chain

The capstone wires together everything from earlier modules: Pythia8 generates events, writes them to HepMC; Geant4 reads that file, propagates particles through the B4 calorimeter, records deposits. You then analyse the ROOT output to measure sampling fraction.

2. Pythia configuration

```
#include "Pythia8/Pythia.h"
#include "Pythia8Plugins/HepMC2.h"

int main() {
    Pythia8::Pythia py;
    py.readString("Beams:ecm = 200.");
    py.readString("HardQCD:hardccbar = on");
    py.readString("PhaseSpace:pTHatMin = 5.");
    py.readString("Tune:pp = 14");
    py.readString("Random:setSeed = on");
    py.readString("Random:seed = 42");
    py.init();

    HepMC::Pythia8ToHepMC toHepMC;
    HepMC::IO_GenEvent out("events.hepmc", std::ios::out);
    for (int i=0; i<1000; ++i) {
        if (!py.next()) continue;
        auto evt = new HepMC::GenEvent();
        toHepMC.fill_next_event(py, evt);
        out << evt;
        delete evt;
    }
}
```

3. Geant4 HepMC reader

Geant4 doesn't ship a HepMC reader by default. Implement a thin `PrimaryGeneratorAction` that reads one `GenEvent` per Geant4 event and transfers final-state particles into `G4PrimaryParticles`:

```
class HepMCGen : public G4VUserPrimaryGeneratorAction {
public:
    HepMCGen(const std::string& fn)
        : fReader(new HepMC::IO_GenEvent(fn, std::ios::in)) {}
    void GeneratePrimaries(G4Event* ev) override {
        auto gen = fReader->read_next_event();
        if (!gen) { G4RunManager::GetRunManager()->AbortRun(); return; }
        auto vtx = new G4PrimaryVertex({0,0,0}, 0);
        for (auto it=gen->particles_begin(); it!=gen->particles_end(); ++it) {
            auto p = *it;
            if (p->status() != 1) continue;
            auto mom = p->momentum();
            auto g4p = new G4PrimaryParticle(p->pdg_id(),
                mom.px()*GeV, mom.py()*GeV, mom.pz()*GeV);
            vtx->SetPrimary(g4p);
        }
        ev->AddPrimaryVertex(vtx);
        delete gen;
    }
private:
    HepMC::IO_GenEvent* fReader;
};
```

4. Wiring into B4a

Replace B4a's B4PrimaryGeneratorAction with your HepMCGen (pass the HepMC filename in the constructor). Keep everything else. Rebuild.

5. Running the chain

```
# Stage 1: generate events
./pythia_writer

# Stage 2: Geant4 reads HepMC
./exampleB4a -m run.mac
# run.mac: just /run/initialize and /run/beamOn 1000
# the number of events must not exceed HepMC file size
```

6. Expected size and time

Stage	1,000 events
Pythia8 (charm)	~20 s, ~50 MB HepMC
Geant4 B4a	~10 min, ~100 MB ROOT
Analysis macro	~5 s

7. Analysis

```
void cap_analysis() {
    TFile f("B4.root");
    TTree* t = (TTree*) f.Get("B4");
    TH1F* hf = new TH1F("f", "sampling fraction", 50, 0, 0.3);
    double Eabs, Eact;
    t->SetBranchAddress("Eabs", &Eabs);
    t->SetBranchAddress("Eact", &Eact);
    for (Long64_t i=0; i<t->GetEntries(); ++i) {
        t->GetEntry(i);
        double tot = Eabs + Eact;
        if (tot > 10) hf->Fill(Eact / tot);
    }
    hf->Fit("gaus");
}
```

}

8. Deliverables checklist

Deliverable	Criterion
events.hepmc	≥ 1000 events, inspect first event text
B4.root	N_entries equal to events processed
Fit f	Mean 0.10–0.20, $\sigma/\mu \leq 15\%$
Plot	f distribution with Gaussian fit overlay
Written summary	1 paragraph: config, stats, result

9. Extensions to consider

- Swap in HIJING events instead of Pythia (use hmain.f with an ASCII dump and a simple reader on the Geant4 side).
- Change the B4 geometry to 70% Pb / 30% scintillator and repeat.
- Add a second SD for a hadronic calorimeter behind the EMCal and measure total-energy resolution.
- Write the full four-vector ntuple (not just Eabs, Eact) and reconstruct jets.

10. Recurring pitfalls

Issue	Resolution
Geant4 exits after one event	HepMC file reached EOF; reduce <code>beamOn</code> count
Sampling fraction = 0	Wrong logical volume is sensitive, or <code>Eact</code> not filled
σ/μ is huge ($> 30\%$)	Beam is off-axis or beyond calorimeter coverage — check direction
Only photons show up	Pythia decays are off; enable 111:onMode etc. or accept $\pi^0 \rightarrow \gamma\gamma$